



H2S

# BHARATIYA ANTARIKSH HACKATHON 2026



Team Name : GARUDA\_AI

Team Leader Name : SOGALAPALLI MANJUNATH REDDY

Problem Statement : Route Resilience: Occlusion-Robust Road Extraction & Graph-Theoretic Criticality Analysis for Urban Mobility

## Team Members

Team Leader:

Name: Sogalapalli Manjunath Reddy

College: Keshav Memorial Institute of Technology

Team Member-1:

Name: Anthati Nikhil

College: Keshav Memorial Institute of Technology

Team Member-2:

Name: Gaurav Pendyal

College: Keshav Memorial Institute of Technology

Team Member-3:

Name: Perma Kushal Madhan

College: Keshav Memorial Institute of Technology

# 03

## OPPORTUNITY / USP

THE PROBLEM WE SOLVE. THE IMPACT WE DELIVER.

**PROJECT GARUDA**  
Zero-Isolation. Always.



### THE REALITY: CHENNAI FLOODS (2023 CASE STUDY)

Extreme rainfall + vulnerable infrastructure =  
rapid isolation of critical services



### WHY EXISTING SYSTEMS FALL SHORT



#### REACTIVE APPROACH

Alerts come after isolation has already occurred



#### NO CONNECTIVITY INTELLIGENCE

Cannot predict when & where isolation will happen



#### DATA SILOS

Fragmented data, no fused real-time intelligence



#### NO ACTIONABLE PRIORITIZATION

Limited ability to optimize resources & routes

### THE OPPORTUNITY: PREDICT. PREVENT. PROTECT.

#### PROJECT GARUDA ZERO-ISOLATION INTELLIGENCE



**GARUDA**  
Zero-Isolation. Always.

#### PREDICT

Time-to-Isolation (TTI)



Predicts isolation hours to days in advance with high spatio-temporal

#### UNDERSTAND

Digital Twin & City Pulse



Real-time digital twin of Chennai to monitor risk, infrastructure health & city pulse

#### FORESEE

Risk & Percolation Simulation



Simulates flood spread & infrastructure failure to identify vulnerable zones & critical cuts

#### ACT

Zero-Isolation Engine



Recommends optimal interventions, alternate routes & resource pre-positioning

#### LEARN

Adaptive Model Learning



Continuously learns from new data & outcomes to improve future predictions

### THE IMPACT



#### ZERO ISOLATION OF HOSPITALS

No critical services left behind



#### SAVE LIVES & REDUCE LOSSES

Faster response, lower human & economic cost



#### RESILIENT & PREPARED CITIES

Data-driven decisions for a climate-resilient future



#### SCALABLE NATIONWIDE IMPACT

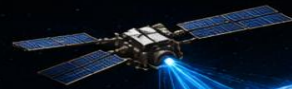
Replicable framework for all Indian cities

**FROM REACTIVE TO PROACTIVE.**  
**FROM ISOLATION TO RESILIENCE.**



GARUDA transforms fragmented data into predictive, actionable intelligence to ensure **Zero-Isolation** of lives, infrastructure & essential services.





# PROJECT GARUDA

## 50 TOTAL INTEGRATED INTELLIGENCE CAPABILITIES

ENGINEERED FOR RESILIENCE INTELLIGENCE • ONE MISSION: **ZERO-ISOLATION**

### ONE MISSION: **ZERO-ISOLATION.**

No Indian community should become digitally invisible before, during, or after a disaster.



### 12 SIGNATURE INNOVATIONS (What makes GARUDA unique)

#### 01 TIME-TO-ISOLATION (TTI)

Predicts how long specific villages / communities will remain reachable before they become isolated for action.

#### 02 GATEKEEPER NODE RANKING

Identifies & ranks transport critical bridges & roads whose failure can accentuate flood & fragmentation.

#### 03 CITY PULSE SCORE

Real-time resilience score reflecting the city's operational health & transport network status.

#### 04 RESILIENCE DIGITAL TWIN

Live digital replica of the transport network for continuous simulation, impact analysis and planning.

#### 05 NEXT FAILURE PREDICTION

Predicts the infrastructure component most likely to fail next with confidence & risk assessment.

#### 06 DISASTER SCENARIO SIMULATION

Simulates floods, landslides & wind scenarios to understand future impacts & vulnerabilities.

#### 07 INFRASTRUCTURE ACCESSIBILITY

Assesses real-time accessibility of critical infrastructure and evacuation routes across the network.

#### 08 HISTORICAL DISASTER LEARNING

Learns from past disaster patterns to improve prediction accuracy and decision support.

#### 09 INFRASTRUCTURE MEMORY

Maintains long-term structural history using repeated satellite & IoT observations learned from the past.

#### 10 GROUND TRUTH INTEGRATION

Integrates crowd reports, IoT & official sources for accurate & trusted decisions.

#### 11 CRITICAL FACILITY ANALYSIS

Analyzes hospitals, schools, shelters & essential services to ensure continuity of operations.

#### 12 CLOSED-LOOP RESILIENCE OS

Continuously optimizes prepare → act → learn cycle for adaptive resilience.

### 38 ADVANCED INTELLIGENCE MODULES (High value, sector-specific capabilities)

#### PERCEPTION & EO INTELLIGENCE (7)

- 01 Optical SAR Fusion
- 02 Multi-Temporal Image Fusion
- 03 Flood Confidence Mapping
- 04 Cloud & Shadow Filtering
- 05 Occlusion Handling
- 06 Change Detection (CD)
- 07 Infrared Hotspot & Fire Detection

7 MODULES

#### HAZARD & ENVIRONMENT INTELLIGENCE (6)

- 08 Rainfall Nowcasting
- 09 DGM-Based Hazard Mapping
- 10 Hydrodynamic Flood Modeling
- 11 Risk & Vulnerability Alerts
- 12 Wind & Cyclone Path Tracking
- 13 Land Use Change Detection

6 MODULES

#### RESILIENCE & IMPACT ANALYTICS (7)

- 14 Resilience Indexing
- 15 Critical Facility Analysis
- 16 Population-at-Risk Estimation
- 17 Evacuation Corridor Planning
- 18 Infrastructure Vulnerability
- 19 Transportation Accessibility
- 20 Scenario Impact Evaluation

7 MODULES

#### DECISION & OPERATION (6)

- 21 Priority Reorder Ranking
- 22 Resource Allocation Support
- 23 Rescue Route Optimization
- 24 Logistics Coordination Engine
- 25 Real-time Incident Support
- 26 Recovery Progress Tracking

6 MODULES

#### INTERFACE & DELIVERY (6)

- 27 Interactive GIS Layers
- 28 Alert & Notification System
- 29 GeoJSON / API Export
- 30 Mobile & Web Access
- 31 Role-based Dashboards
- 32 Multi-language Support

6 MODULES

#### ADAPTIVE & LEARNING SYSTEMS (6)

- 33 Continuous Validation
- 34 Adaptive Learning Loop
- 35 Infrastructure Memory Updates
- 36 Truth Validation Engine
- 37 Graph-based Integration
- 38 Crisis Pattern Recognition

6 MODULES

**OPEN SOURCE TECHNOLOGIES**  
Python, PyTorch, GDAL, SNAP, NetworkX, Leaflet

**ZERO INFRASTRUCTURE COST**  
Runs on Free Cloud / Colab / Kaggle GPU

**DESIGNED FOR IMPACT**  
Scalable • Transparent • Actionable for Decision Makers

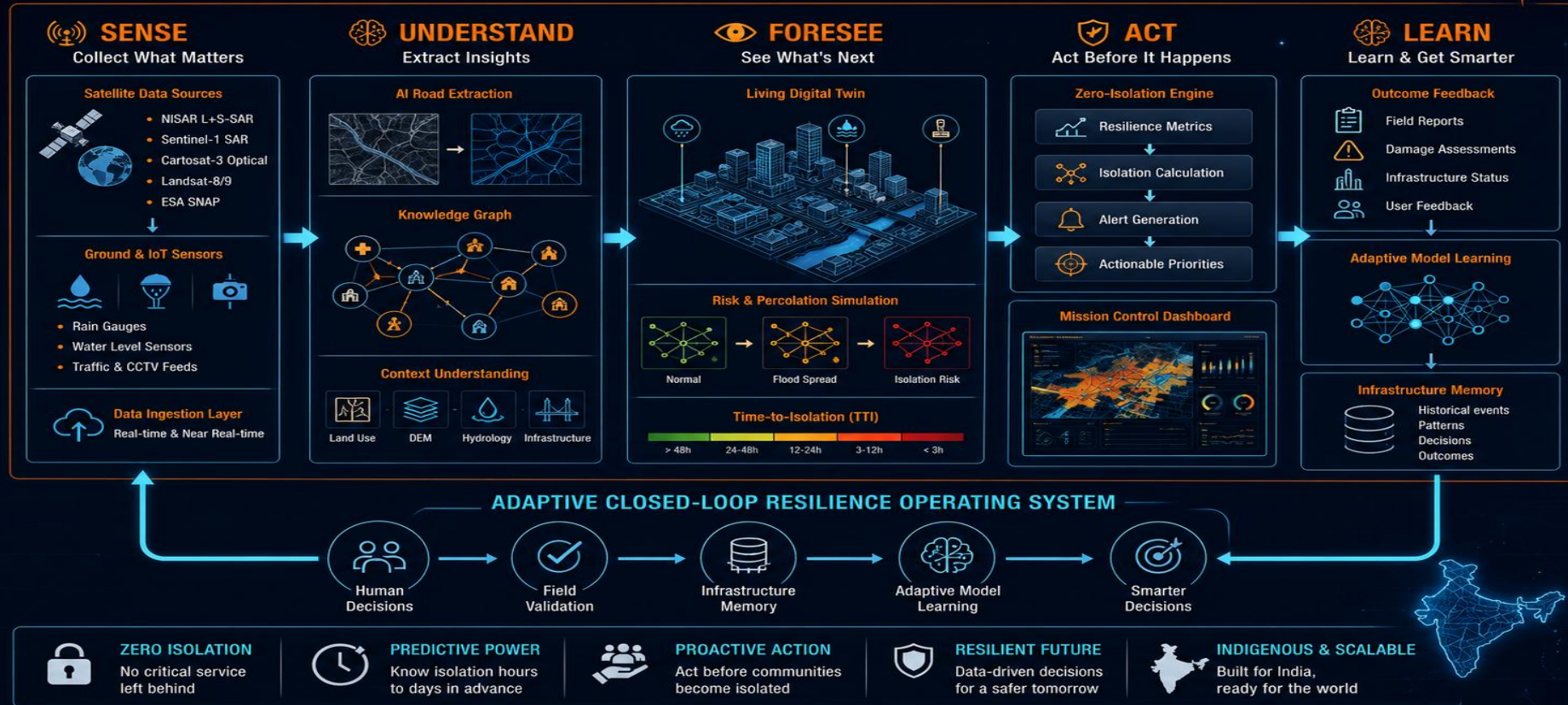
**PURPOSE: ZERO-ISOLATION**  
Resilient Communities. Connected India. Stronger Nation.

“ No Indian community should become digitally invisible — before, during, or after a disaster.”

**GARUDA** turns India's satellites into a disaster-ready nervous system for its cities.

# PROJECT GARUDA

## SYSTEMS ARCHITECTURE: END-TO-END OPERATIONAL FLOW



SENSE. UNDERSTAND. FORESEE. ACT. LEARN. For a Zero-Isolation India.





# PROJECT GARUDA

## MISSION CONTROL DASHBOARD

Real-time Resilience Intelligence Command Center

28 JUN 2025  
11:42:35 IST

SYSTEM OPERATIONAL



Overview

City Pulse

Risk Map

Infrastructure

Alerts

Digital Twin

TTI Monitor

Analytics

Reports

Settings



RESILIENCE INDEX

76 / 100

High Resilience

↑ 8% vs yesterday



CITY PULSE SCORE

68

Moderate

↑ 5% vs yesterday



TIME-TO-ISOLATION (TTI)

42 min

Estimated

↓ 15 min vs yesterday



ACTIVE ALERTS

7

High Priority

[View All](#)



AFFECTED POPULATION (EST.)

2.48 L

↑ 11% vs yesterday



WEATHER NOW

26°C

Heavy Rain

Wind: 18 km/h

Humidity: 88%

RESILIENCE RISK MAP (Real-time)

RISK LEVEL

- Very High
- High
- Moderate
- Low
- Very Low

MAP LAYERS

- Flood Risk
- Infrastructure
- Critical Facilities
- Transport Network
- Water Bodies



REAL-TIME ALERTS & NOTIFICATIONS

[View All](#)

- Bridge Overtopping Risk** 11:41 IST  
Adyar Bridge  
Water level rising fast. Overtopping possible in 30-45 min.
- High Flood Risk Zone** 11:39 IST  
Okkium Thoraiakkam  
Accumulated rainfall high. Flash flood possible.
- Traffic Disruption Likely** 11:37 IST  
GST Road (Chrompet)  
Waterlogging detected. Slow traffic expected.
- Rainfall Intensity High** 11:35 IST  
South Chennai  
Heavy rainfall continuing. Monitor closely.

7 Active Alerts

[View All](#)

RESILIENCE DIGITAL TWIN

3D CITY MODEL



[View 3D](#)

[Layers](#)

[Simulate](#)

ACTIVE MISSION

CHENNAI FLOOD WATCH

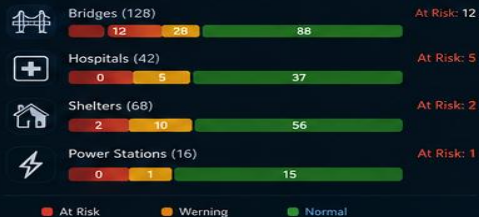


Data Source  
CARTOSAT-3 +  
RISAT-1A + GPM

Last Updated  
11:42 IST

CRITICAL INFRASTRUCTURE STATUS

[View All](#)



GATEKEEPER NODES (Top 5)

Why Important?

| Node                | Criticality Score | Impact if Fails | Status  |
|---------------------|-------------------|-----------------|---------|
| Adyar Bridge        | 96                | Very High       | At Risk |
| Kathipara Junction  | 91                | Very High       | At Risk |
| Manali New Town     | 79                | High            | Warning |
| Velachery Main Road | 72                | High            | Warning |
| Poonamallee High Rd | 68                | Moderate        | Normal  |

[View All Gatekeeper Nodes](#)

AI RECOMMENDED ACTIONS

Powered by GARUDA AI

- Monitor Adyar Bridge continuously.** Priority: HIGH  
Overtopping probability high in next 45 min.
- Pre-position NDRF teams in Okkium area.** Priority: HIGH  
High flood risk & evacuation probability.
- Implement traffic diversion on GST Road.** Priority: MEDIUM  
Waterlogging likely in next 30-60 min.
- Alert nearby hospitals & shelters.** Priority: MEDIUM  
Ensure readiness for incoming cases.

[View Action Plan](#)

DATA SOURCES



CARTOSAT-3  
Optical Imagery



RISAT-1A  
SAR Data



GPM IMERG  
Rainfall Data



IMD  
Weather Data



Census India  
Population Data



OpenStreetMap  
Base Map

LAST DATA UPDATE

28 JUN 2025, 11:42 IST



**GARUDA** – Turning India's Satellites into a Disaster-Ready Nervous System for its Cities

# PROJECT GARUDA

## ADAPTIVE CLOSED-LOOP RESILIENCE OPERATING SYSTEM

### REAL-TIME RESILIENCE STATUS

|                      |             |
|----------------------|-------------|
| RESILIENCE INDEX     | 0.65        |
| CURRENT DANGER       | 0.38        |
| ACTIVE TTI ALERTS    | 3           |
| HOSPITALS AT RISK    | 3           |
| BHOONIDHI API ACCESS | ₹0 OVERHEAD |

### MULTI-SOURCE SENSING

- SAR & Optical Data
- IoT & Ground Sensors
- Rain, Water Level, Traffic & CCTV
- Crowd & Field Inputs

### ACTION & RESPONSE

- Zero-Isolation Engine
- Actionable Alerts
- Resource & Route Optimization
- Mission Control Dashboards



L-BAND SAR  
(NISAR)

S-BAND SAR  
(Sentinel-1)

OPTICAL /  
MULTISPECTRAL  
(Cartosat-3)

**ZERO-ISOLATION. ALWAYS.**

No Indian community should become digitally invisible before, during, or after a disaster.

### POWERED BY EARTH OBSERVATION

NISAR  
(L-band SAR)

Sentinel-1  
(S-band SAR)

Cartosat-3  
(Optical)

Bhoonidhi  
APIs



**1 SENSE**  
Collect  
What Matters



**2 UNDERSTAND**  
Extract  
Insights

### INTELLIGENCE LAYER

- AI Road & Feature Extraction
- Knowledge Graph & Context
- Risk & Vulnerability Assessment
- Pattern Recognition



**4 ACT**  
Act Before  
It Happens



**3 FORESEE**  
See  
What's Next

### PREDICTIVE INTELLIGENCE

- Flood & Percolation Simulation
- Time-to-Isolation (TTI) Prediction
- Critical Infrastructure Risk Forecasting
- Scenario & What-If Analysis



**5 LEARN**  
Learn &  
Get Smarter

## ADAPTIVE CLOSED-LOOP RESILIENCE ENGINE

Continuous sensing, intelligence, action and learning for a Zero-Isolation India.

### ADAPTIVE LEARNING FEEDBACK LOOP



**ZERO ISOLATION**  
No critical services  
left behind



**FASTER RESPONSE**  
Hours to days  
in advance



**RESILIENT & PREPARED**  
Cities stay prepared,  
communities stay safe



**REDUCED LOSSES**  
Save lives & reduce  
economic impact



**SUSTAINABLE & SCALABLE**  
Built for India,  
ready for the world



### GARUDA MISSION

No Indian community should become digitally invisible before, during, or after a disaster.

**Zero-Isolation. Always.**

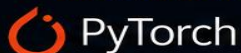


# PROJECT GARUDA: TECHNOLOGY ECOSYSTEM & FRUGAL IMPLEMENTATION MATRIX



## 1. MACHINE LEARNING CORE

### CORE FRAMEWORK



### KEY LIBRARIES

- SegFormer / Mask2Former (Vision Transformers)
- Albumentations (Data Augmentation)
- OpenCV (Image Processing)
- NumPy (Numerical Computing)
- Pandas (Data Handling)
- Scikit-learn (ML Utilities)

### PURPOSE



AI models for flood mapping, road risk prediction and decision intelligence.



## 2. GEOSPATIAL & EARTH OBSERVATION DATA



BHOONIDHI API ACCESS  
₹0 OVERHEAD

### DATA SOURCES & TOOLS

- Cartosat-3 (High Resolution)
- NISAR (L+S-SAR) (All-Weather SAR)
- ESA SNAP (Multi-Temporal Radar)
- GDAL / Rasterio (Geospatial Processing)
- ISRO Bhoonidhi Open APIs
- Sentinel-1 / Landsat Imagery (via Bhoonidhi)
- OpenStreetMap (OSM) (Base Map & Roads)

### PURPOSE



Multi-source EO data for real-time situational awareness, monitoring and analysis.



## 3. NETWORK TOPOLOGY & SIMULATION CORES

### CORE TOOLS



### KEY MODULES

- NetworkX (Graph Analytics)
- OSMnx (Road Network Extraction)
- Betweenness Centrality
- Percolation Failure Modeling
- Combinatorial Optimization
- Greedy Search Algorithms
- StaMPS InSAR (Deformation Analysis)
- Time-Series Trend Analysis

### PURPOSE



Understand network vulnerabilities, simulate failures and identify critical infrastructure risk.



## 4. FRONTEND, DATA & MANAGEMENT INTERFACES



MISSION CONTROL UI

### TECHNOLOGIES

- FastAPI (Asynchronous Backend)
- Streamlit (Dashboards)
- PostGIS (Spatial Database)
- NDEM (Interoperable Formatting)
- React.js (UI Components)
- Leaflet.js (Interactive Maps)
- GitHub (Version Control & Collaboration)

### PURPOSE



Seamless data management, visualization and mission decision support.



# TOTAL FOUNDATIONAL SETUP OVERHEAD: ₹0 (NO ADDITIONAL INFRASTRUCTURE COST)

|                                                                                   | RESOURCE SECTOR                                               | FINANCIAL BURDEN       | UNDERLYING SYSTEM REASONING                                                                                                                                                                           |
|-----------------------------------------------------------------------------------|---------------------------------------------------------------|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | <b>SATELLITE DATA ASSETS</b><br>Earth Observation Data Access | <b>₹0</b><br>ZERO COST |  Accessed through ISRO <b>Bhoonidhi</b> and Copernicus open-data services.                                          |
|   | <b>COMPUTE OPERATIONS</b><br>Training & Inference Resources   | <b>₹0</b><br>ZERO COST |  System training optimized on free tier <b>Google Colab</b> and <b>Kaggle Notebook</b> GPU resources.               |
|  | <b>SOFTWARE LICENSING</b><br>Open-Source Ecosystem            | <b>₹0</b><br>ZERO COST |  Engineered entirely using <b>open-source</b> packages and frameworks.                                              |
|   | <b>ENGINEERING CAPITAL</b><br>Human Expertise & Development   | <b>₹0</b><br>ZERO COST |  Developed by a <b>four-member</b> interdisciplinary student team using existing institutional computing resources. |



Prototype developed entirely using publicly accessible datasets, open-source software, and existing academic computing resources.  
**Operational deployment can leverage existing ISRO/NRSC infrastructure.**



**OPEN DATA**  
ISRO Bhoonidhi,  
Copernicus



**OPEN SOURCE**  
No licensing cost,  
maximum flexibility



**EXISTING ACADEMIC RESOURCES**  
Institutions, labs  
& cloud platforms



**NO ADDITIONAL PROTOTYPE COST**  
100% cost-efficient  
development

**ZERO INFRASTRUCTURE INVESTMENT. MAXIMUM IMPACT. BUILT FOR INDIA.**





BHARATIYA  
ANTARIKSH  
HACKATHON

2026

THANK YOU